

# Exploring Virtual Object Translation in Head-Mounted Augmented Reality for Upper Limb Motor Rehabilitation with Motor Performance and Eye Movement Characteristics

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## ABSTRACT

Head-mounted augmented reality (AR) technology is currently employed in upper limb motor rehabilitation, and the degrees of freedom (DoF) of virtual object translation modes become critical for rehabilitation tasks in AR settings. Since motor performance is the primary focus of motor rehabilitation, this study assessed it across different translation modes (1DoF and 3DoF) via task efficiency and accuracy analysis. In addition, eye movement characteristics were used to further illustrate motor performance. This research revealed 1DoF and 3DoF modes showing their own benefits in upper limb motor rehabilitation tasks. Finally, this study recommended selecting or integrating these two translation modes for future rehabilitation task design.

## CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; User studies.

## KEYWORDS

Head-mounted augmented reality, Translation modes, Rehabilitation, Motor performance, Eye movement

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## 1 INTRODUCTION

The popularity of augmented reality (AR) technology has made it easier for patients to receive effective motor rehabilitation outside

clinics nowadays [1]. AR rehabilitation systems combine virtual objects with physical world, thereby connecting patients' virtual and actual experiences. Due to the AR advantages, patients can receive multisensory feedback and conduct the training more effectively in motor rehabilitation.

The main focus of motor rehabilitation is to enhance patients' motor performance, which refers to the motor learning outcome of rehabilitation training [2]. Previous literature indicated that motor performance, such as task efficiency and precision, is associated with skill acquisition in rehabilitation [2]. Besides, it is reported in the literature that people's eye movement presents its potential to explore motor performance in visual tasks [3].

Researchers have developed an AR rehabilitation system, which allows patients with upper limb motor deficits to move virtual objects as part of their rehabilitation training [4]. Virtual object translation, which is often applied in rehabilitation training, becomes one of the fundamental aspects in the AR environment [5]. However, two widely adopted modes for object translation, degrees of freedom (DoF) separation and DoF integration, are seldom explored in AR rehabilitation [6]. Specifically, 1DoF mode based on the concept of DoF separation refers to the mode of limiting virtual object translation on the x-, y-, or z-axis separately, whereas 3DoF mode based on the concept of DoF integration refers to the mode of moving virtual object along all three axes arbitrarily [7].

In order to simulate upper limb motor rehabilitation training in AR, this study created some virtual object translation tasks with different translation modes in the head-mounted AR setting. This study aims to explore the effects of two virtual object translation modes (1DoF and 3DoF) on motor performance along with eye movement characteristics. The contributions of this study are outlined as follows: (i) evaluating the correlation between virtual object translation modes and motor performance in AR rehabilitation tasks; and (ii) exploring users' motor performance via eye movement characteristics in AR rehabilitation tasks.

## 2 USER STUDY

Ten health participants (six males and four females) aged between 20 and 28 years ( $M=24$ ,  $SD=4.3$ ) were recruited to experience the head-mounted AR rehabilitation tasks in this study. Before the user study, participants were briefed on the overall purpose of the

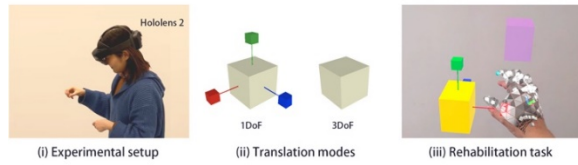
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**Figure 1: Experimental design: (i) experimental setup (ii) two virtual object translation modes, and (iii) rehabilitation task.**

experiment, and then they signed an individual consent form. This research has been approved by the University Ethics Committee.

Figure 1 (i) displays the experimental setup which contained the Microsoft HoloLens 2 (an AR headset) installed with upper limb motor rehabilitation tasks.

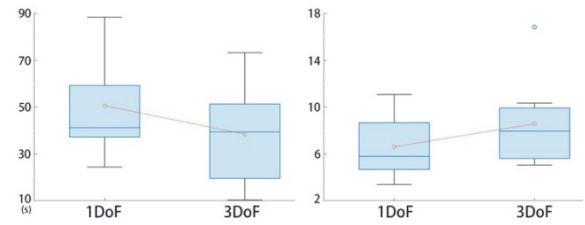
As shown in Figure 1 (ii), participants were required to employ 1DoF and 3DoF modes to complete corresponding tasks. In 1DoF translation mode, participants used mid-air gestures to select one of the three arrows by grasping it, and then moved their hands to translate the virtual object on the x-, y-, or z-axes separately combined with the real world. In 3DoF translation mode, participants used mid-air gestures to grasp the virtual object, and then moved their hands to translate the virtual object on all three axes in synchrony with the physical surroundings.

Before the experiment, participants had chance to adapt to these two translation modes. Then, they were given to complete three tasks for each translation mode. In each task as displayed in Figure 1 (iii), one moveable object and one target position were showed in the combined virtual and real environment. Participants were asked to translate the moveable object to the target position as precisely as possible without time constraints. In the experiment, participants could walk around in a 2.5m×2.5m physical space while wearing the AR headset.

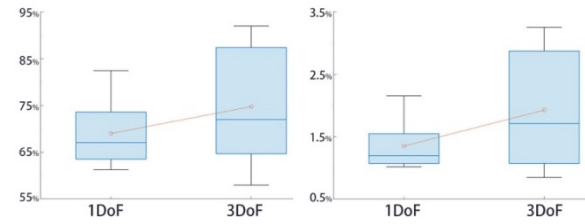
Measures, including motor performance and eye movement, in each task were recorded by the AR program for further analysis. The manipulation time and error rate of each translation task were collected for motor performance analysis. Eye movement were tracked by eye tracker in the HoloLens 2 with sampling rate of 30 Hz. The fixation length ratio (the ratio of time spent focused on the moveable object to the total manipulation time) and the first fixation ratio (the ratio of the time to initially gaze at the moveable object to the total manipulation time) were employed for eye movement analysis. The Wilcoxon signed-rank test was used to assess statistically significant variations in these measures across translation modes.

### 3 RESULTS

As shown in Figure 2, no significant variations were observed in manipulation time between two translation modes, but participants spent less manipulation time in 3DoF mode ( $M=38.427$ ) than in 1DoF mode ( $M=50.617$ ) on average. Besides, although there were no significant differences in error rate across two translation modes, error rates were slightly lower in 1DoF mode ( $M=6.613$ ) than in 3DoF mode ( $M=8.578$ ) on average.



**Figure 2: Motor performance: completion time (left) and error rate (right).**



**Figure 3: Eye movement characteristics: fixation length ratio (left) and first fixation ratio (right).**

As displayed Figure 3, there were no significant variations in the fixation length ratio between two translation modes, but the fixation length ratio was higher in 3DoF mode ( $M=0.748$ ) compared with 1DoF mode ( $M=0.690$ ) on average. On the other hand, although no significant differences were found in the fixation length ratio between two translation modes, the first fixation ratio was slightly lower in 1DoF mode ( $M=0.014$ ) than 3DoF mode ( $M=0.019$ ) on average.

### 4 DISCUSSION

Previous research has revealed that the major emphasis of motor rehabilitation training is motor performance [2]. This research compared two indexes of motor performance, i.e., task efficiency and accuracy, of two object translation modes in AR. 3DoF mode was more efficient and required less time to complete translation tasks on average. However, in 1DoF mode, participants were able to translate the virtual object to the target position with greater precision on average. Hence, each of the two translation modes may have its own benefits and drawbacks in rehabilitation tasks.

Prior research revealed that eye movement characteristics correspond with motor performance [3]. This research adopted two eye movement characteristics to further illustrate motor performance in rehabilitation tasks. Based on the means of the fixation length ratio, participants kept longer visual attention on virtual objects when adopting 3DoF mode, which may result in greater task efficiency. Based on the means of the fixation length ratio, participants paid slightly sooner attention to virtual objects in 1DoF mode, which may have been positively correlated with task precision in this experiment.

Based on motor performance and eye movement analysis, this research recommends selecting or combining 1DoF and 3DoF modes to enhance task efficiency and precision owing to their distinct benefits in AR rehabilitation tasks. Particularly, 3DoF mode may

be a better option for motor rehabilitation, especially when the patients lack the ability to control the object along one axis due to motor dysfunction and the quick completion of rehabilitation tasks is beneficial for patients. As 3DoF mode allows users to conduct rehabilitation tasks more quickly, they could complete more rehabilitation training in less time, leading to enhanced skill acquisition. Furthermore, 3DoF mode lacks unnecessary visual information, allowing users to focus more on the virtual object which they were currently controlling. However, this study indicates that 1DoF mode may be a feasible option if more precise rehabilitation training is desired. In addition, 1DoF mode could be conspicuous and attract users' attention at first glance. Hence, this study suggests 3DoF mode as the primary option and 1DoF mode as the secondary option for upper limb motor rehabilitation tasks.

## 5 CONCLUSION

This study focuses on users' motor performance along with eye movement characteristics with 1DoF and 3DoF translation modes in AR rehabilitation tasks. 3DoF mode was related to enhanced task efficiency with the greater fixation length ratio, while 1DoF mode contributed to increased task precision with the lower first fixation ratio. Finally, this research suggested to select or integrate these two translation modes for further upper limb motor rehabilitation task design.

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## REFERENCES

- [1] Ying, W. and Aimin, W. Augmented Reality Based Upper Limb Rehabilitation System. 2017. In 2017 13th IEEE International Conference on Electronic Measurement & Instruments (ICEMI), pages 426–430. IEEE. <https://doi.org/10.1109/ICEMI.2017.8265843>
- [2] Wang, L., Huang, M., Wang, Y., Yang, R., Liao, K. L., Zhang, J., and Sun, W. 2021. Movement Modulation in Virtual Rehabilitation: its Influence on Agency and Motor Performance. In 2021 IEEE 9th International Conference on Serious Games and Applications for Health (SeGAH), pages 1–8. IEEE. <https://doi.org/10.1109/SEGAH52098.2021.9551897>
- [3] Matessa, M. 2004. Anticipatory Eye Movements in Interleaving Templates of Human Behavior. In 2004 International Conference on Cognitive Modeling, IH-058.
- [4] Höll, M., Oberweger, M., Arth, C., and Lepetit, V. 2018. Efficient Physics-Based Implementation for Realistic Hand-Object Interaction in Virtual Reality. In 2018 IEEE Conference on Virtual Reality and 3D User Interfaces (VR), pages 175–182. IEEE. <https://doi.org/10.1109/VR.2018.8448284>
- [5] Goh, E. S., Sunar, M. S., and Ismail, A. W. 2019. 3D Object Manipulation Techniques in Handheld Mobile Augmented Reality Interface: A Review. IEEE Access, 7, 40581–40601. <https://doi.org/10.1109/ACCESS.2019.2906394>
- [6] Sun, W., Huang, M., Yang, R., Han, J., and Yue, Y. 2021. Mental Workload Evaluation of Virtual Object Manipulation on WebVR: An EEG Study. In 2021 14th International Conference on Human System Interaction (HSI), pages 1–6. IEEE. <https://doi.org/10.1109/HSI52170.2021.9538783>
- [7] Sun, W., Huang, M., Wu, C., and Yang, R. 2022. Sense of Agency on Handheld AR for Virtual Object Translation. In 2022 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW), pages 838–839. IEEE. <https://doi.org/10.1109/VRW55335.2022.00270>